Balancing King

Game Design Document

NOTE: THIS DOCUMENT IS CURRENTLY INCOMPLETE

Sections with an asterisk (*) are missing content.
Content marked with dashes (-) indicates what should be added.
Text in Purple indicates new mechanics that are yet to be officially applied

Text in Red indicates Tweaks that are yet to be applied

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Game Overview

*Game Concept

A platformer/rage game where the player balances an egg on their head while navigating through different traps and obstacles. The player can tilt their head to counterbalance movement and is at constant risk of the egg falling off. Some areas may require a bit of puzzle-solving to navigate through.

*Genre

2.5D puzzle platformer / Physics-driven

A very minor overarching story and difficulty curve should be steady.

*Target Audience

Casual players who enjoy quirky mechanics and challenges. Players who enjoy games with high Skill Floors and even higher Skill Ceilings, where progression is not through levels but mastering the mechanics to reach further on the next attempt.

*Core Pillars of Design

The player would think about watching their speed and adjusting how they move so the king doesn't fall.

The player should feel tense when the king starts wobbling, but relieved when they recover it.

The player should balance risk and safety by slowing down when possible and speeding up when needed.

Easy to Learn / Hard to Progress

Overall experience, basic movement, should have a degree of challenge that requires experience and mastery to consistently perform. New players should encounter immediate challenges while still having a feeling of progression / learning from each failure. Expert players should be able to overcome these challenges with relative ease.

Constant Risk / Balancing Mechanic

The game imposes a constant risk of failure and loss of progression if the player is reckless. The balancing mechanic should always be at the back of the player's mind and be a factor in every obstacle design going forward.

Fairness / Character Control

Players should never feel 'cheated' by losses (deaths should be justified). Every obstacle should be react-able and have a consistent 'counterplay' (even if the counterplay is difficult). The game should not expect hardcore precision platforming. Player patience should be rewarded (allowing them to stop between obstacles to figure out how to approach it).

Controlling the character, even if difficult, should be consistent and refined. Physics of the egg and character behaviours should never suddenly switch, and there should be little to no randomness.

Unique / Interesting Core Mechanic (the EGG)

Movement throughout the level should have some level of uniqueness to standard platformers and challenge the player in creative/interesting ways to usual platformers. These mechanics should be robust and expandable to allow for many fun situations.

Punishment / Progression and Learning from Mistakes

To keep the player tension high, punishments should be harsh and cause the player to replay all (or part) of the level. Replaying parts of the level should still be a challenge and encourage mastery that helps with future, more difficult challenges.

The player should be rewarded (i.e. dialogue, visual changes) for reaching milestones in the level to keep the intrigue and motive of reaching the end.

*Game Flow Summary

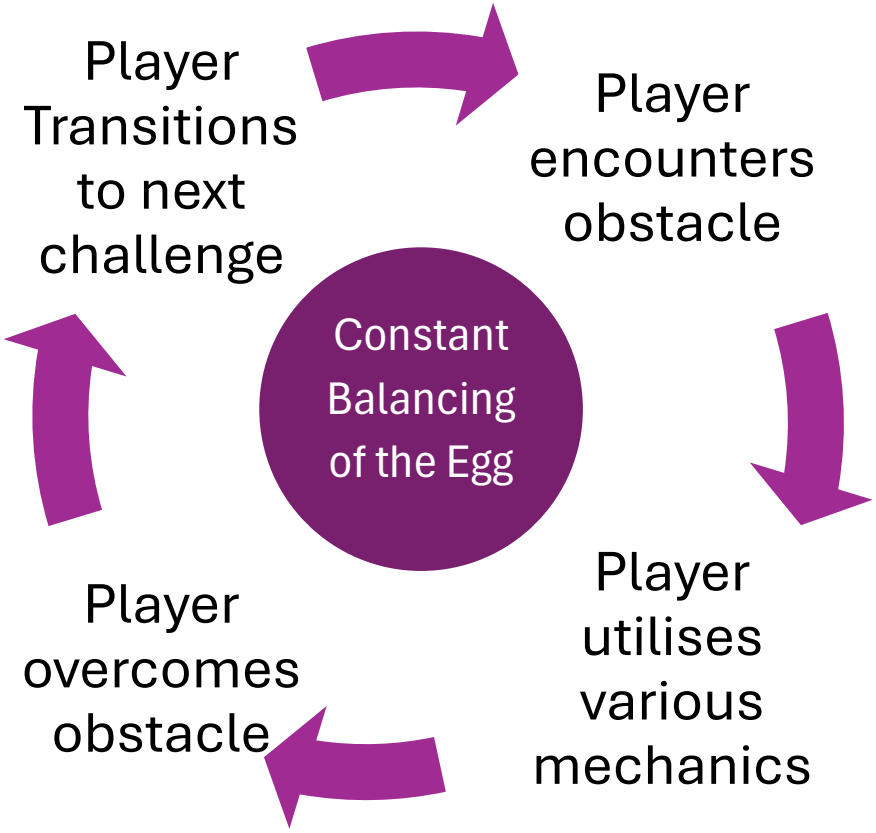
The game will be a **Linear Level** that is around **15 minutes Long.** Within this level there are obstacles (either individually or in groups) that will aim to tilt the player off balance and force them to drop the egg.

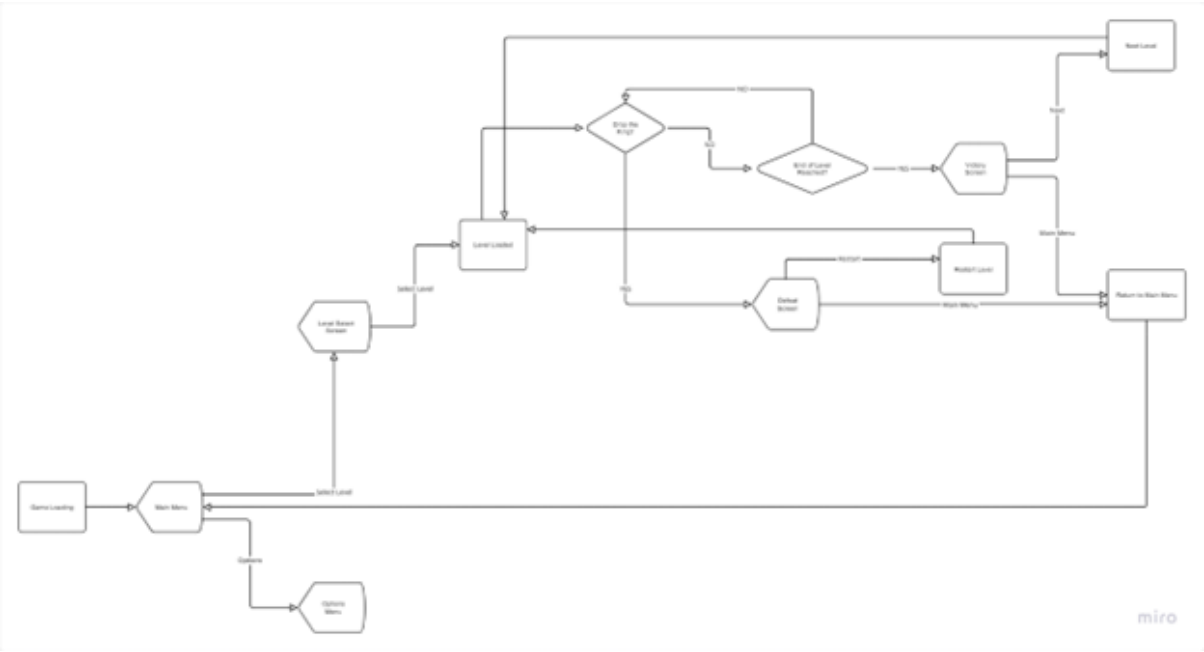
The player wins if...

They navigate successfully to the end of the level without dropping the egg.

The player can lose by...

Dropping the egg and having it break.





*Look and Feel

- Colourful, silly, stylised world
- Comedic elements with wobbly physics and animations
- Somewhat frustrating but a generally light-hearted tone

*Inspiration

- Human: Fall Flat (comedic physics-based game)
- Getting Over It (slow-paced, linear level design)

Gameplay and Mechanics

*Player Character Mechanics

Player Character Gameplay Design

The player character balances a “basket” on their head that contains the egg. The egg inside is loose and is physics-based, meaning heavy/reckless movements can cause it to spill out onto the ground.

New Basket – Bowl

The player character has a bowl / egg-cup style basket so there is equal weighting in both sides, leading to more flexible level design in the future. It will also be more punishing due to a less steep back

Player Character Movement Design



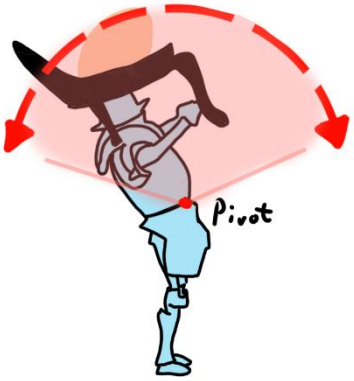
When the character moves, there should be an added risk of the egg wobbling and falling off. This should deter reckless movement and be an added challenge for new players. Experienced players should be able to move quickly and carefully.

New Leaning Mechanic

The player now leans

Player Tilting

The player can also use the controller to adjust the tilt of the “basket” left and right by pressing the LT and RT triggers of the controller. The controller will have haptics so the player can adjust the angle based on how firmly they press the triggers down, giving them a lot of control. The character tilts on a pivot that is on their hip.



Player Jumping

The player can press a button to jump, allowing them to jump over obstacles.

Luke Designer Notes: Currently, the only time you need to jump is for the rolling barrels so it doesn’t get used much at all; beyond that, it doesn’t really add anything to encourage the player to use it. Maybe we can add more traps that get more use out of it! Furthermore, it doesn’t really have any uses beyond that, maybe another mechanic to make it more advantageous to the player (maybe launch the egg to hit a button or something)?

Player Crouching

The player can crouch, allowing them to duck under objects.

Luke Designer Notes: Once again, it is only used for one trap and 1 function. Much like the jump, it could have something small tied into it (less slippery or something) that gives it more appeal!

Luke’s Current Character Thoughts

I feel that although there is a base balancing mechanic to make things more tense, there is nothing beyond that which allows the player to master it / there isn't much depth in it at its current form. I think maybe we should look into adding a mechanic that gives the balancing/egg a lot more depth and expression, so we can have more creative challenges than just: walk->trap->walk->trap. In games

like Getting Over it, the movement is very robust (lets the player climb, launch, hang) with the actual mountain challenging the player to master this movement. I think we could maybe try and add something to give the player (and us) a bit more to work with!

Some ideas could be: **Launching the egg, More control over balancing, more in-depth impact of jumping and crouching...**

Ben's Notes: Like Luke mentioned, some extra movement or mechanic ideas would help the game feel less linear and have more depth to master. A **charge jump** could expand movement and level design boundaries, though that might only work if we include vertical movement and level sections. Any "charge" must take in account that the egg is a separate entity and applying that force from the knight may make the physics act weird in certain situations.

Maybe **jumping can increase the player's speed**, leading to a need to master jumping and catching the king again. – for speedruns if we also include a timer.

Or a different mechanic where you must **get the king somewhere or through something** to advance the level, using the crouch or jump features outside of avoiding traps.

Some sections of the map may require the player to crouch to navigate through tighter spaces, effectively linking the level design with a player mechanic.

*Obstacles / Puzzles

All traps will be activated and deactivated by triggers when the player collides with them. When the player enters the activation trigger, the trap will begin to run its function, with some repeating in a cycle and others having a single action cycle. When the player passes through the deactivation trigger, the trap will finish and reset.

For any obstacles, these will be baked into the environment to be a hazard for the player. They will have to be careful when getting past them, as the obstacle will affect their balance.

Ben's Notes: The current system for the traps needs to be fully reworked so they can act more robustly and can be easily expanded for further traps. For any traps that use the player's jumping and crouching ability, a lot of testing and fine-tuning may need to be done.

- Each trap can have their activation and deactivation traps individually, or a group of them can have a single trigger for the activation. This needs to be discussed.

Each trap could also have variants, like barrel traps which are a single barrel or another which has bursts of barrels coming at the player, as an example.

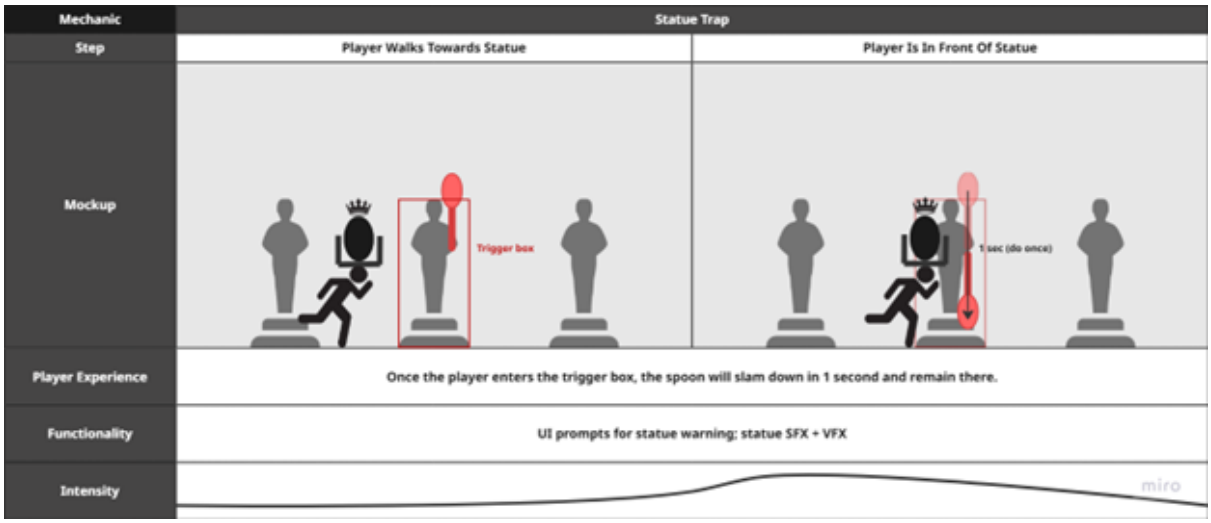
Luke's Notes: I agree with Ben!

Statue Trap

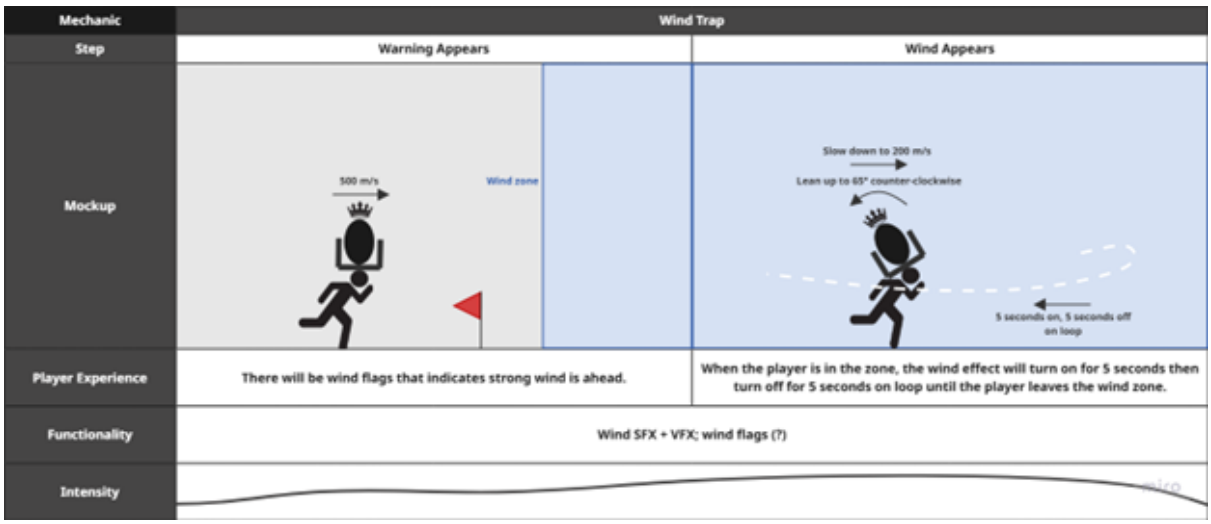
When activated, the spoon will rotate downwards quickly, waiting for a small time, before slowly returning to its original position. This will repeat after a small cooldown, continuing its cycle.

Once the player has passed, they will go through another trigger which will deactivate the statue trap.

Luke Notes: Right now, the spoons kill the player if they touch it (even if they keep the egg balanced). This feels really jarring. I feel maybe the only way the player should lose is if they drop the egg, so maybe give this a rework

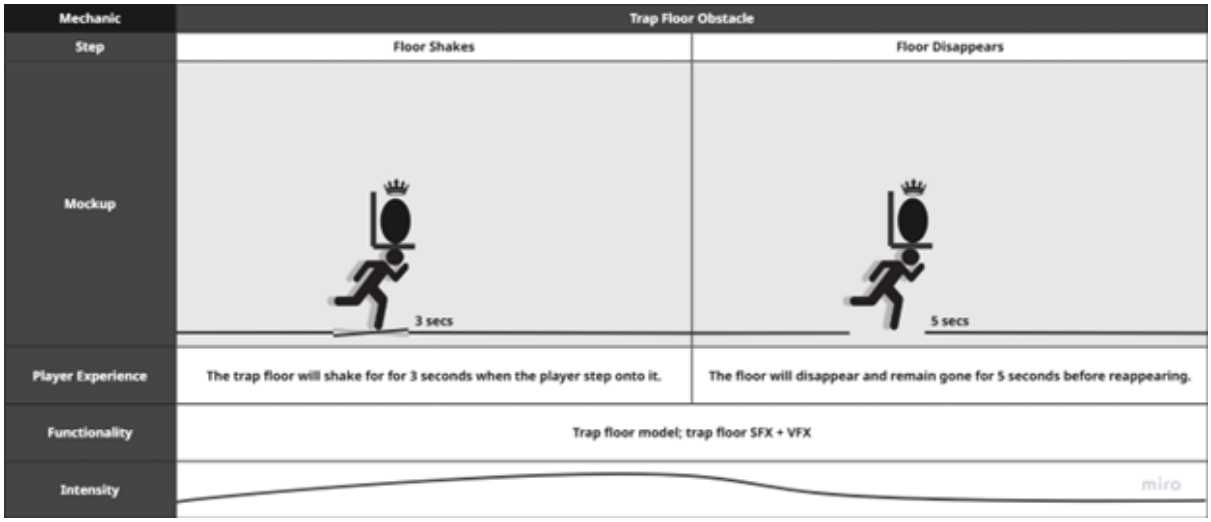


Wind Obstacle



Ben's Notes: This is not currently in the level but have blueprints in the files to be adapted. Could be a nice addition to an environmental obstacle for the player to battle against.

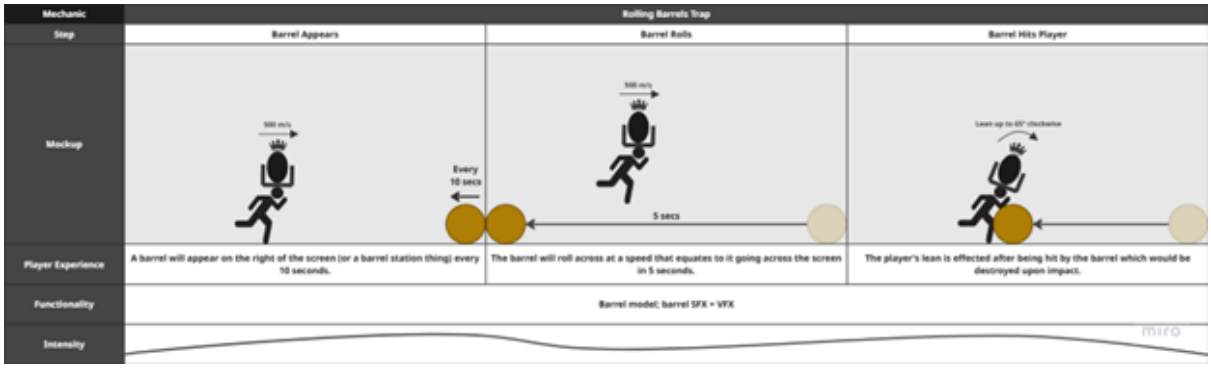
Trap Floor



Ben's Notes: Currently is not within the project at all, from what I've seen, and could pose an issue for the level. Will need to be discussed.

Rolling Barrels

Luke's Designer Notes: Currently, the barrel is fast and has NO indication of coming. This makes it quite unreadable, which leads to frustrating deaths. They feel very difficult to jump over and appear during areas where you are supposed to crouch (putting you in a catch-22)!



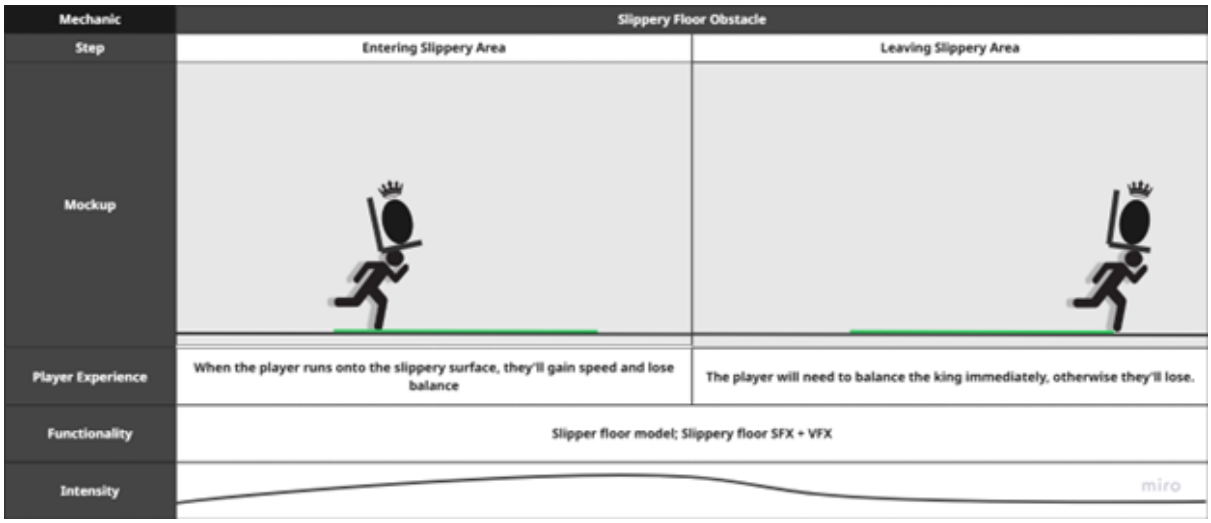
When inactive, the barrel mesh is invisible and not interactable with. When the player runs through the activation trigger, the barrel comes into existence and

rolls towards the player. With enough warning, the player must jump over and catch the king on the other side.

If the barrel hits the player, an effect on the player's balance is applied and the barrel gets "destroyed" (deactivated in function). However, if the player avoids the barrel, it will deactivate after a small time. Both ways will reset the barrel's position back to the start so it can activate again later.

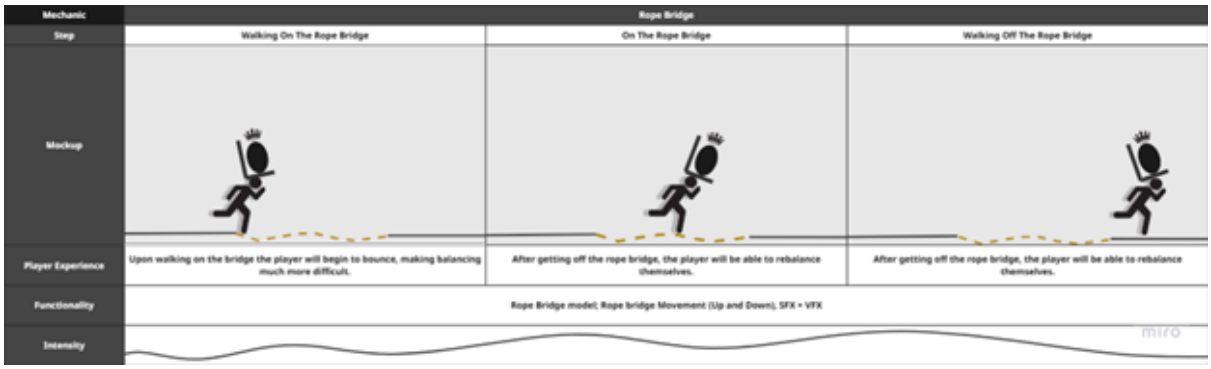
Ben's Notes: The barrel trap currently activates once and then never runs again. Needs to be changed to reset once run, and a variant could be made to make a burst cycle of multiple barrels that the player must avoid.

Slippery Floor



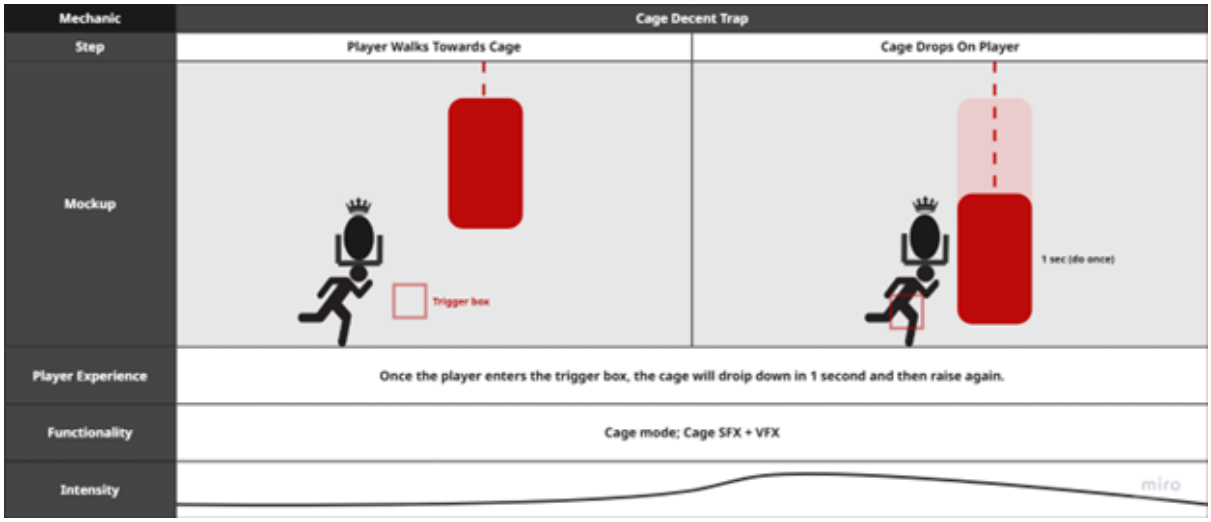
Ben's Notes: Not in the project files, but can easily be implemented with not much pressure on the art and animation side. When the player runs on the more slippery floor, the speed of movement and tilt multiplier will be increased. These will reset back to default values when the player moves off this floor.

Rope Bridge



Ben's Notes: The bridge is in the project and works well. We could do some fine-tuning to the bounciness of the planks, and where it is in the level to make it more sense, but the function overall is good in my opinion.

Cage Decent

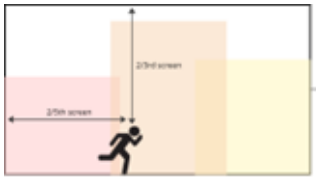


When activated on trigger, the cage will move down from above and block the player's path. The player will have to stop moving before getting squashed. Once the cage is fully down, it will wait a moment before rising back up to give the player a chance to move through. The cage will then repeat the movement after a small cooldown and continue to do so in a cycle. This will repeat until the player passes through and deactivates the trap, making the cage ascend back to its original position and remain there.

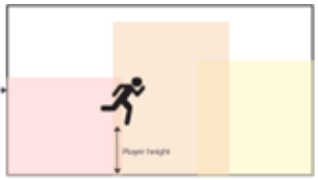
Ben's Notes: This trap currently acts weirdly, not moving all the way down and not presenting much of an issue for the player. It also never resets, remaining in the down position indefinitely. Needs to be changed.

Luke Notes: The positions of these traps in the level combine with certain traps that make it extremely difficult (in an unfair way), e.g. forcing you to crouch then immediately chucking a barrel at you, we should consider this in the future!

*Camera Perspective



- Player will remain at a slight offset to the left so that the player can always see ahead.
- Player will face towards the right regardless of the direction they're going to accommodate for the balancing mechanic.
- Zoomed out camera shows off the environment and enables the player to better see their surroundings.



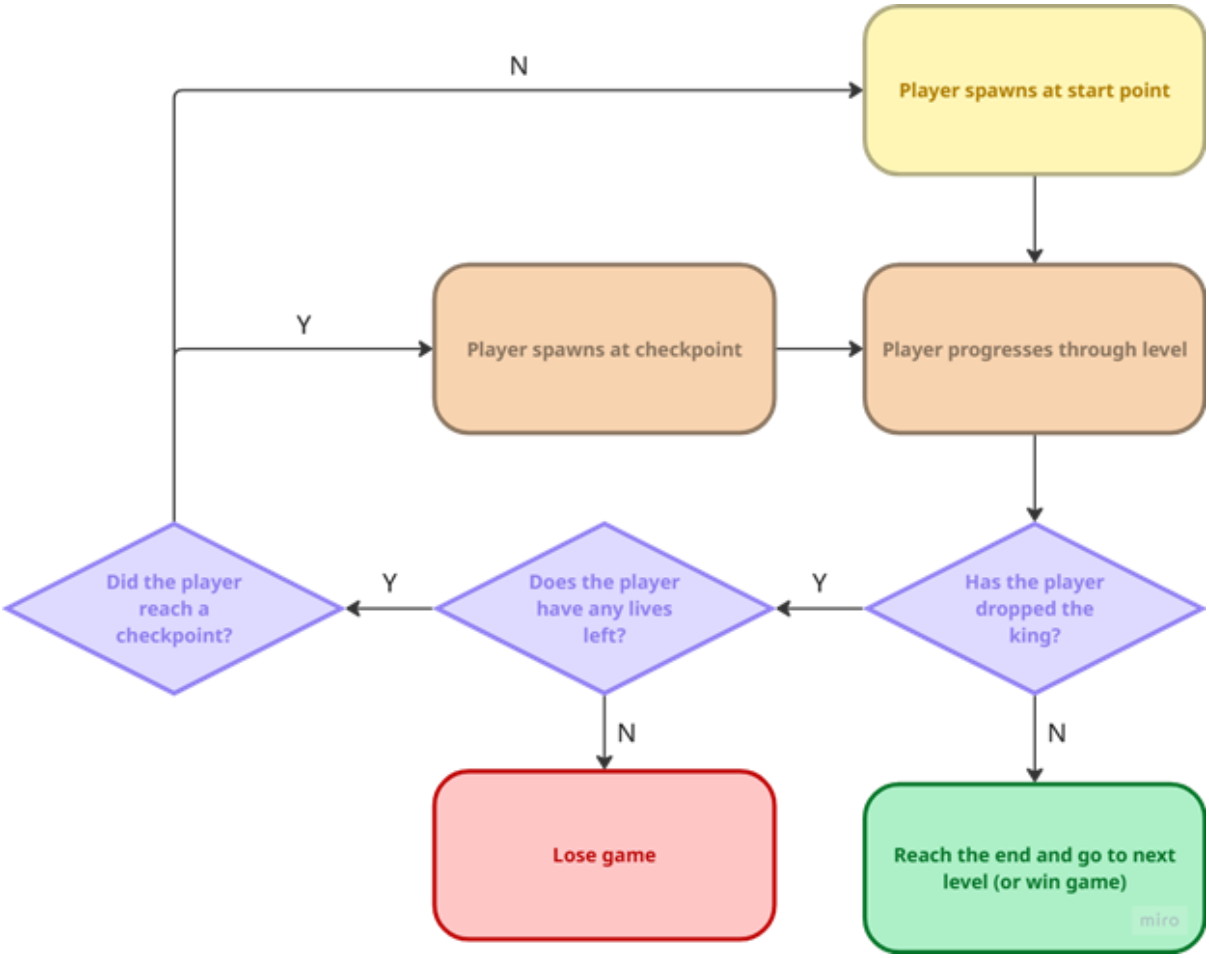
- Player's jump will be as high as their height.
- The camera will not move vertically with the character when jumping.



- There will be a dynamic camera system where the camera on specific occasions will zoom in - adds intensity.

miro

*Player Win & Loss States



Win: Reach the end without breaking the object.

Loss: Object falls off and breaks.

There will be checkpoints and quick restarts to ease frustration.

*Checkpoint System



Ben's Notes: Reduce the number of checkpoints that we have in the game, making it less forgiving to the player when they make a mistake. More of the

checkpoints could be towards the start, being more forgiving when the player is learning how best to play.

Idea from Maria: instead of just walking through the checkpoints, the player must throw the king into a cup to gain the checkpoint.

Brainstorm / Spit ball Ideas!

Mechanics / Gameplay

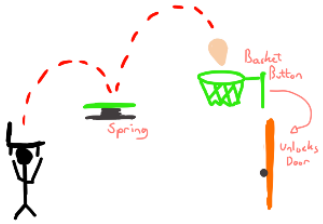
- Instead of the movement setting the tilt target angle to a value, the movement input gets converted into a percentage value for how far the knight should tilt between 0 and the max angle in that direction.



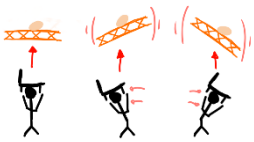
- **Bowl / Nest -** Having a more bowl-like design with a less-deep back and front could prove to be more challenging and invite some creative features, such as walking backwards.
- **Adding a launching the egg feature**
  - Allows for more skill expression in the character movement
  - Launching the egg could allow for more "puzzle-solving" navigation, as well as add more risk for the player catching the egg
  - This could allow for different types of traps and challenges



  - **Example / Potential "Launching" Challenges**
  -
    - **Buttons for doors** that only the egg can reach if launched
    - **Springs-like object** that launches the egg even higher
    - **Crack-proof** floors, allowing for instances where the player must leave the egg and catch it later
    - **See-saw trap** that tilts the way the player is tilting when close enough







  - **Scope considerations:**
    - Current physics could make this have a lot of bugs/inconsistencies that may make it difficult
    - Animations – Could be addressed by slowing down the crouch animation and speeding it up in reverse to launch it

- **Extra Trap Ideas -**
- Swinging axe -

    - Starts swinging on a cycle when activated, can be horizontal or vertical.
- Fire spitting pillars -
    - Player gets a small warning before a fire vfx is fired and the player must avoid the trap.
    - Can be horizontal or vertical, just like the axe.
- Uneven ground -
    - Environmental obstacle which will affect the player's balance.
- Object slamming down – like a Thwomp from Mario.
    - A faster cage descent trap.

Art / Visuals

- Add Dante from Devil May Cry (Example)
- New Character idea: since the background of the game feels quite empty currently, Alex came up with a new idea for a character. We're not going with something very complex, because it'll definitely take longer to make (usual character pipeline that I've been taught was that it takes 6 weeks to create a character) Alex thought that a goblin would look good for the inside of the cells, so depending on what y'all think, I could come up with one or two goblins. Other creature ideas are welcome! They will be posed and I'll try to animate a breathing cycle for them since I've done it before and I think it's quite easy to make. (Alex mentioned a goblin sleeping on the floor, but I also think that I could pose a goblin holding onto the metal bars) In terms of concept art/sketches, I was thinking of a thin character. If we want two, I can definitely make one that's on the bigger side if we go with goblins.
- New prison/cell props: bones and/or skulls
-

Narrative

*The World

[Provide Information]

Kings Castle -> Outside / Village -> Cavern -> Dungeon -> Other castle

*The Characters

*King Eggward VIII

[Provide Information]

*Sir Poachalot

[Provide Information]

*Queen Egglizabeth

[Provide Information]

*Story Arcs

[Provide Information]

Levels

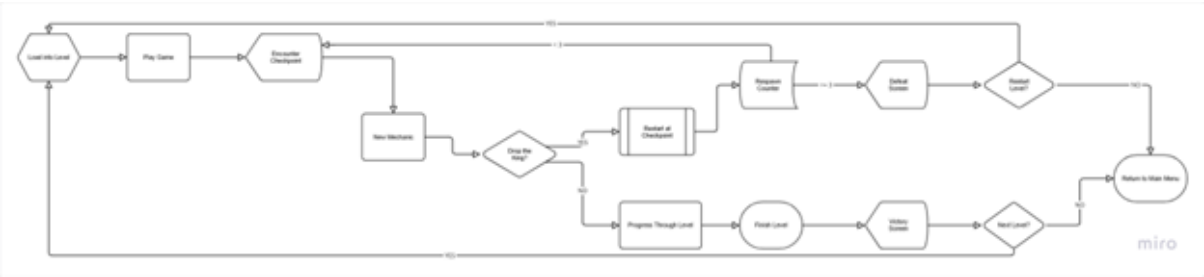
*Design Philosophy

- Always prioritise the balance mechanic
- Introduce hazards by themselves before combining them
- Avoid stinginess with checkpoint placements
- Break up challenging sections with safe zones
- The player will be worse than the game than you think they will be

Ben's Notes: Only 1 level, roughly 15 minutes long. Depending on the skill level of the player, this may be a lot longer or a lot quicker.

*Tutorialisation

Each level should briefly introduce the player to any new mechanics; however, in some cases, this can be avoided if the mechanic is visually simple to understand.



Ben's Notes: We currently have no tutorial in the game. While not directly needed, the player needs to be taught in some way, so they don't feel like they've been cheated by the game system. An idea that I had was to use the level design, and potentially random voice lines if we wanted to, to teach the player. Having the difficulty curving upwards through the level will ensure the starting part is easy enough to learn on the first playthrough. More checkpoints will be available earlier on to help the player, and then they will become less frequent very quickly.

Other ideas from Maria include drawings and funny text boxes from the king to tell the player what to do and where to go. Could give expressions and personality to the king – making the player laugh or making them intentionally hate the king.

*Level 1 (ACTUAL FIRST LEVEL): Eggward's Castle

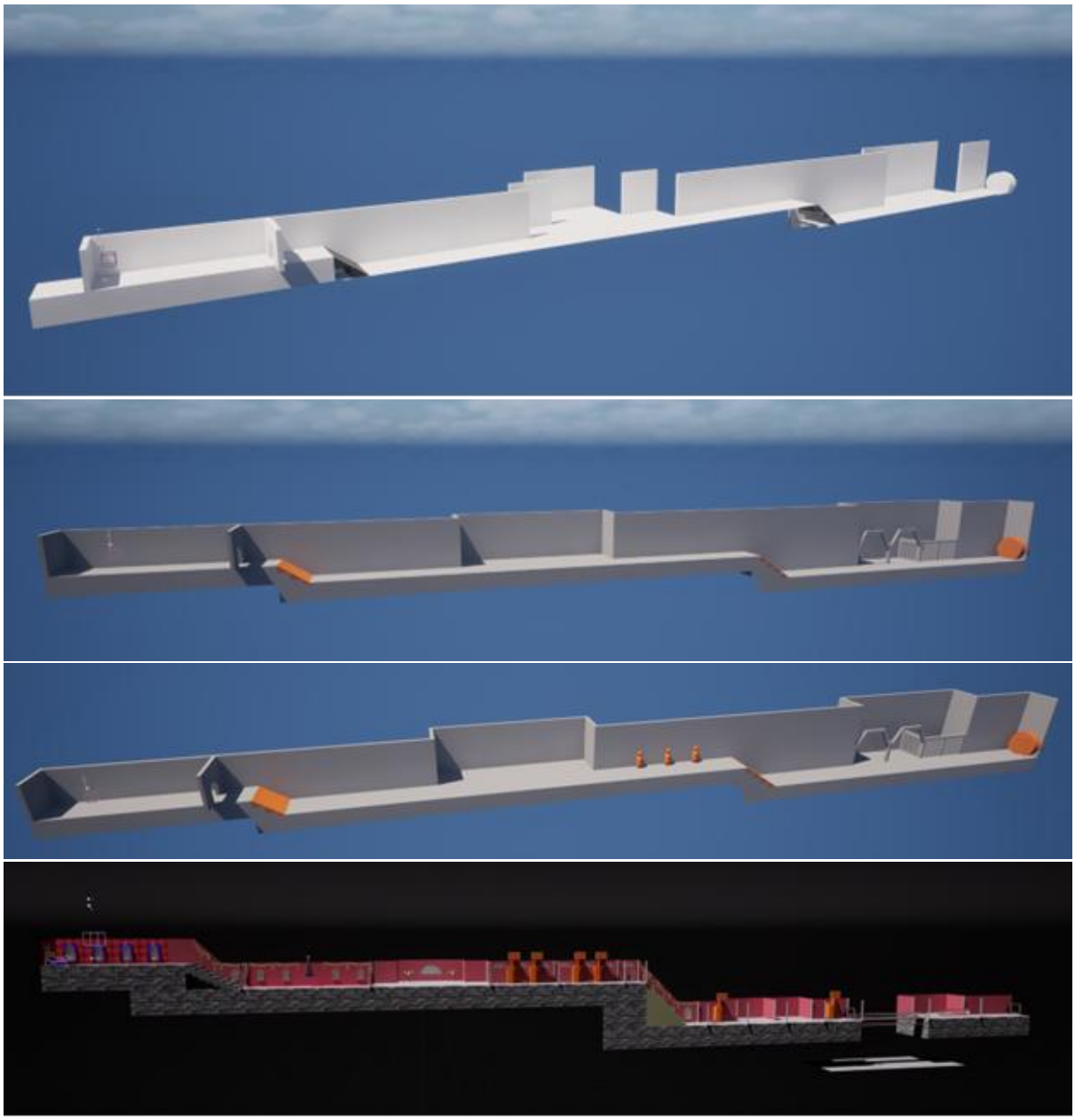
1. Eggwards' Sleeping Quarters
2. Royal Court
3. Dungeon
4. Sewer

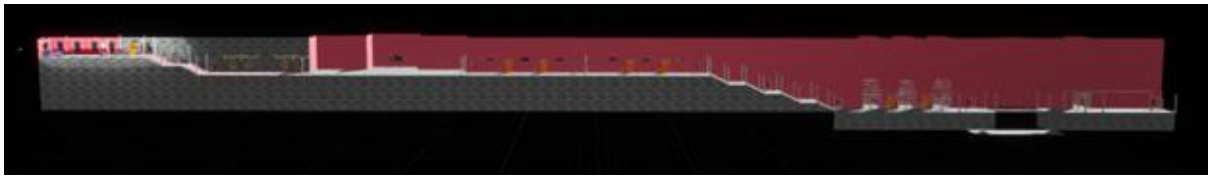
***Development Note**: Sewers in between two levels (just a small hole at the end of the level or something).*

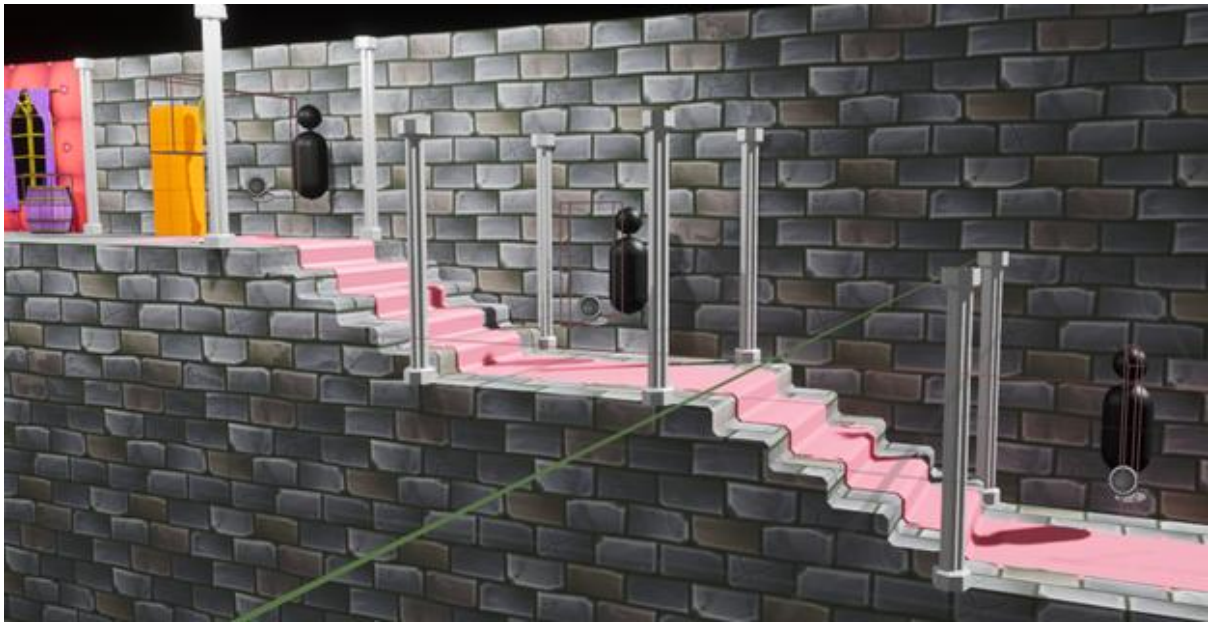
Concept Art of level one.



Blockout(s)









*Level 2: Royal Gardens

1. Sewer outflow (entrance/exit)
2. Village (Crops, occasional buildings)
3. Forest Edge

***Development Note**: ends sneaking in the back of a cart or something.*

*Level 3 (ACTUAL FINAL LEVEL): Princess's Castle

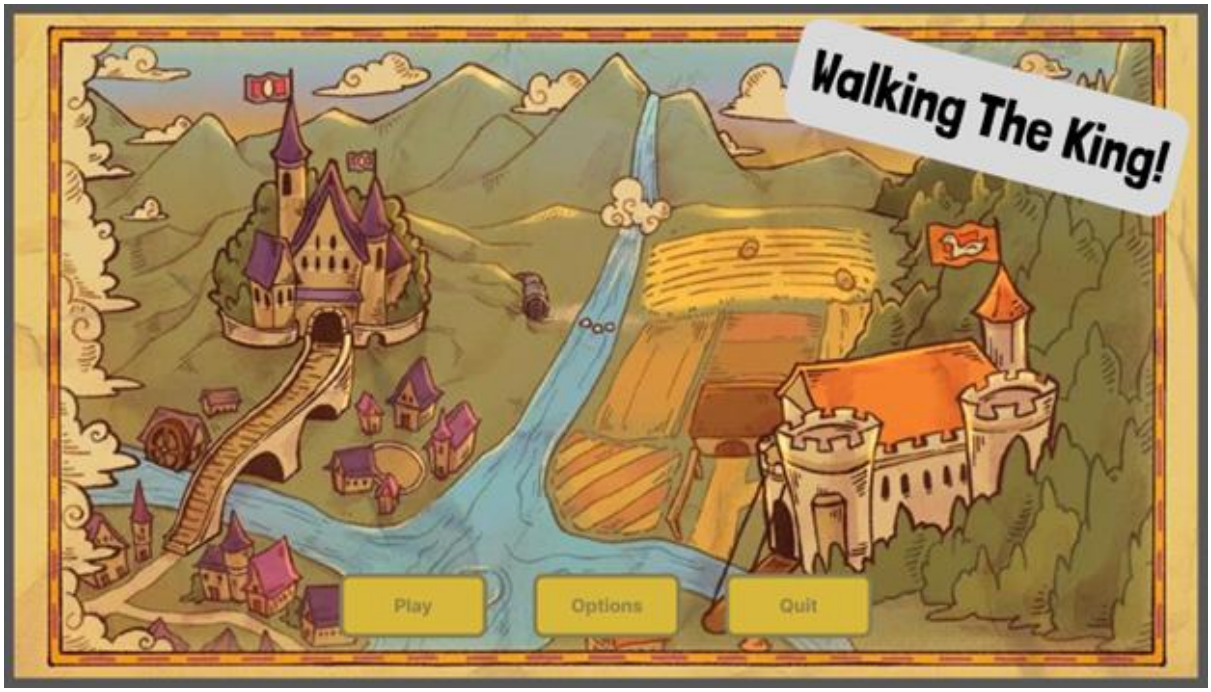
1. Village
2. No Egg’s land (battlefield style)
3. Soldiers' post (Uphill level)
4. Egglizabeth’s Castle

***Development Note**: ends with a black screen showing a few letters written between the two.*

Interfaces

*User Interface

Main Menu



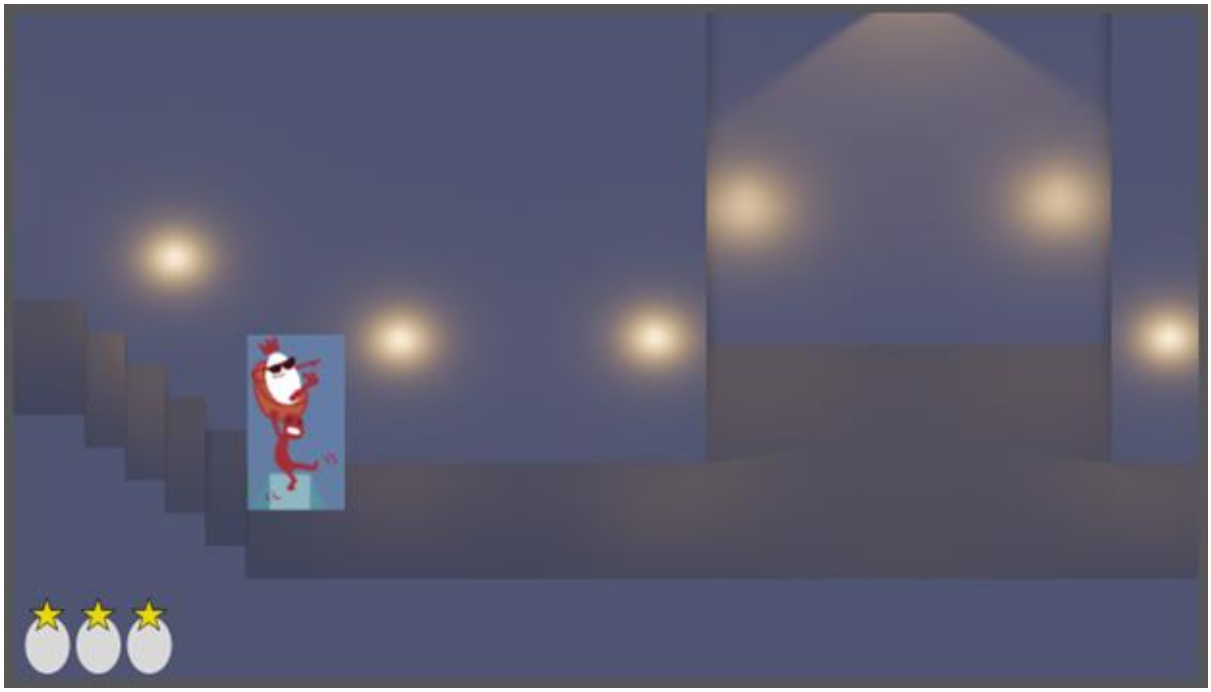
Options / Settings



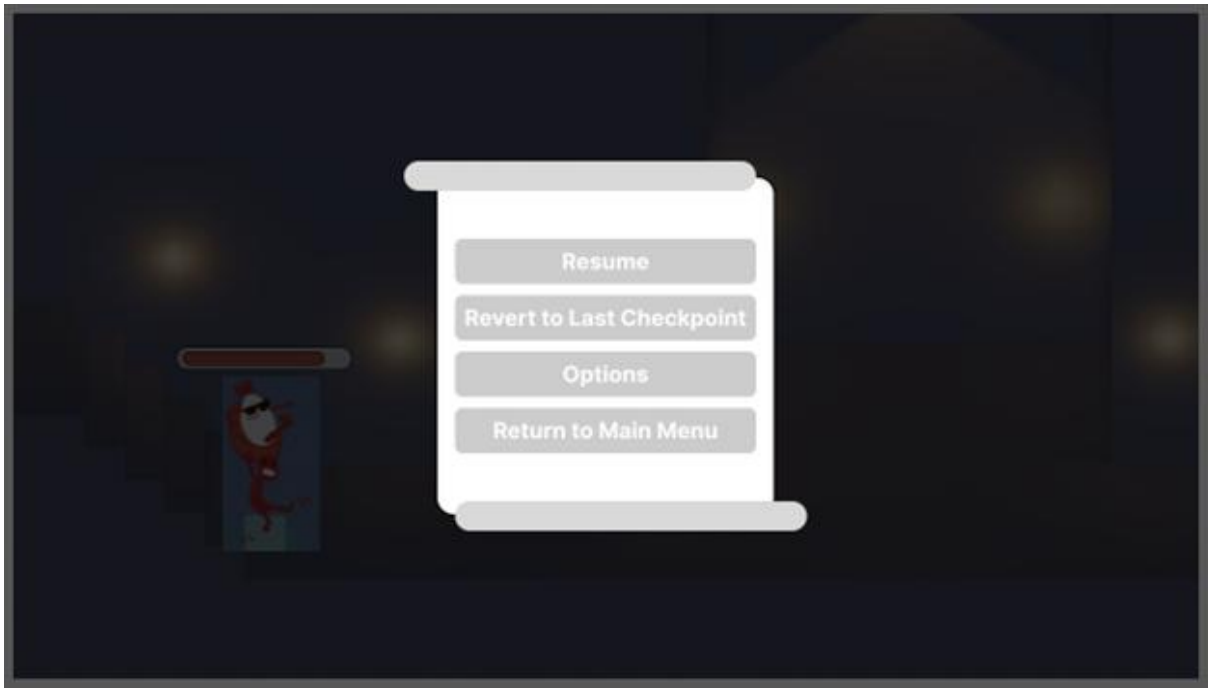
Level Selection



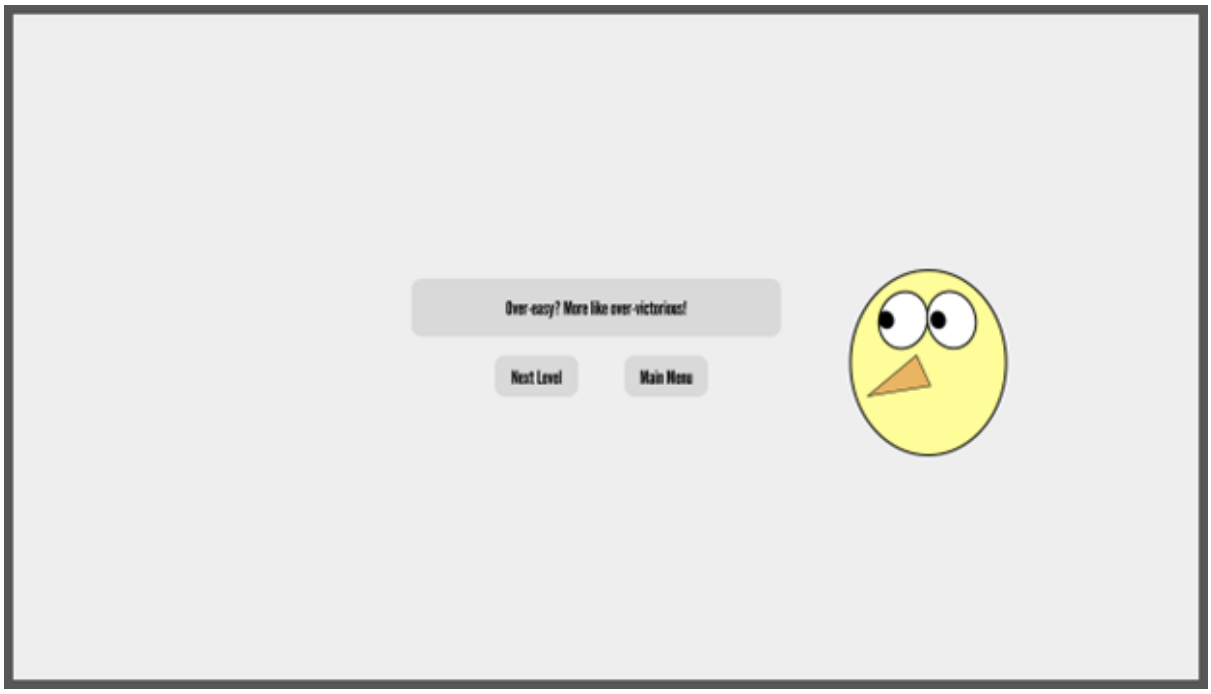
Gameplay HUD



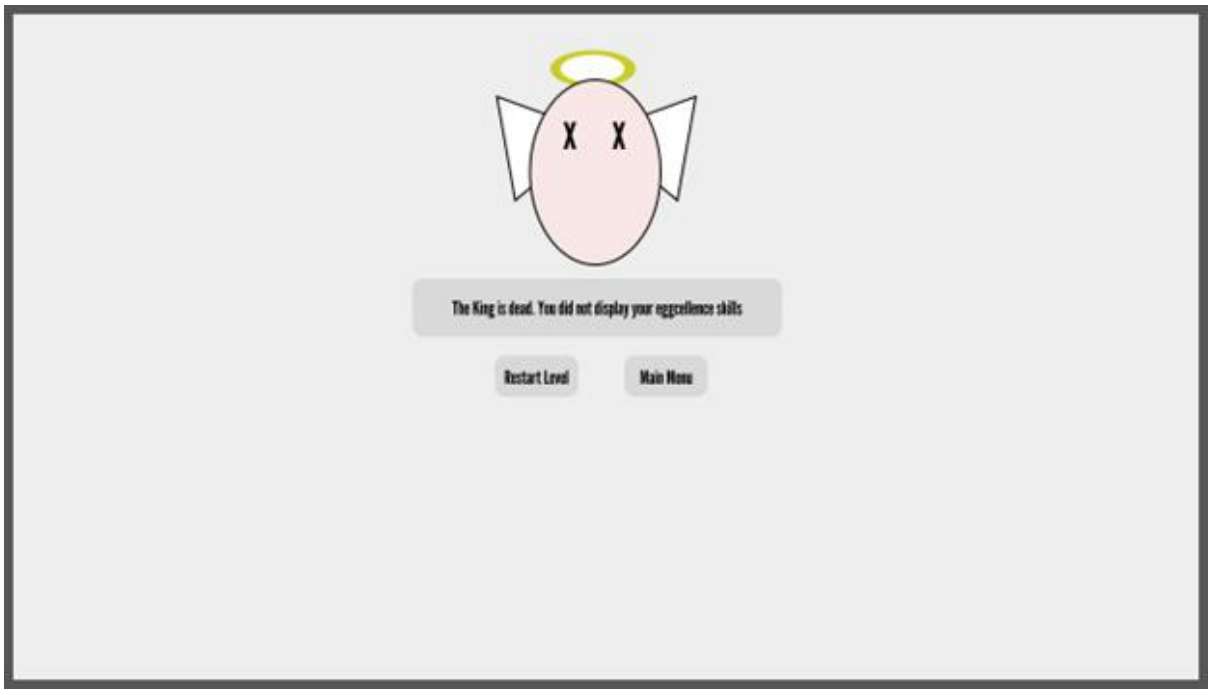
Pause Menu



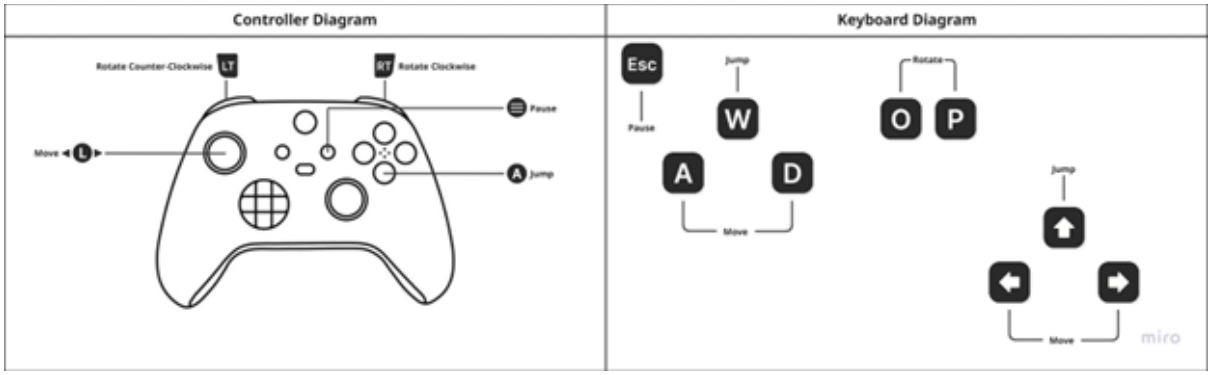
Victory Screen



Defeat Screen



*Control System



The player has to balance the king on the throne he carries via speed; the faster he goes, the king will lean back and slowly go towards falling off. Same with stopping, stopping too suddenly will make him fall.

The balance should have a specific angle where the king will fall, about 65 degrees being the most he can lean.

Jumping should pose a challenge as speed might be high, so the player is incentivised to take it slow.

Audio

*Music

[Provide Information]

All the music and sfx that is needed are in the project, which are all credited from freesound. We may need some more in the future for future mechanics, and we can discuss if any custom or different sound files are wanted to replace what we have.

Audio volume needs to be linked up to the UI sliders in the options screen. This will be linked up through an audio mixer in UE.

*Sound Effects

[Provide Information]

*Voice Acting

Maybe?

[Provide Information]

Animations

I don’t know what to input into this section, as I’ve not mentioned Animation within a GDD without it just being a list of states.

* States

[Provide Information]

Technical Specifications

*Target Hardware

[Provide Information]

*Development Hardware & Software

The following is a list of all specialist hardware and software used during development, along with their method of use during the project.

- GitHub (Used for development of the game in a remote environment)
- Unreal Engine 5.6 (Used for programming and development of the game)
- Blender (Used for creating assets for the game)
- Figma (Used to create UX Wireframes)
- Trello (Organisation / To-Do-List)
- Miro (Moodboards / References / Idea generation)

***To Add**: Expand this list when possible, including licence limitations*

Previous GDD Mechanics / Things

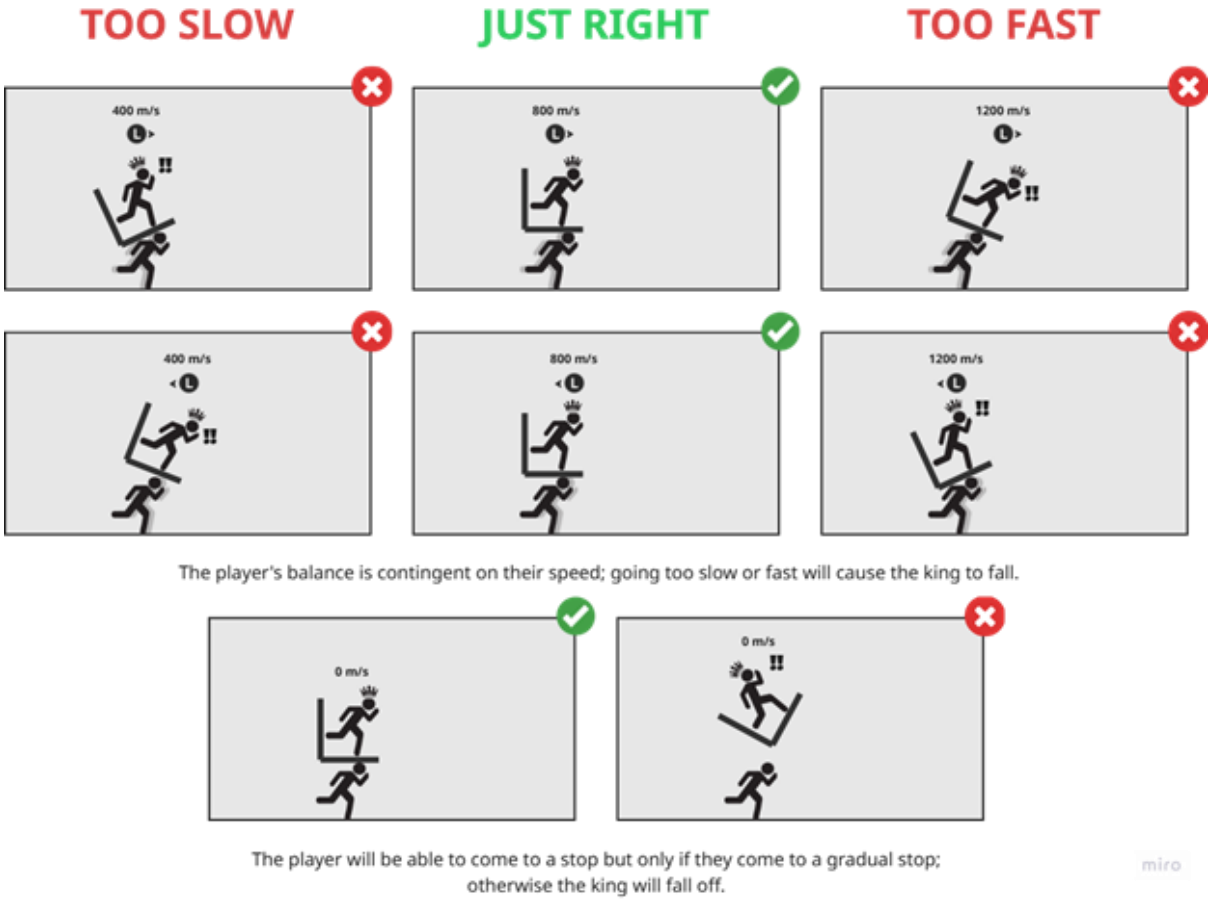
"Basket" - Chair



The player character is currently carrying a throne on its head. This has a backrest on the left side, which allows the player to move quite quickly through the level and adds a little bit of forgiveness to balancing the end. The backrest is quite a sharp corner, which means the player can sometimes run at full speed (not stopping) without the egg coming loose.

Current – Leaning Mechanic

The character tilts its torso based on the player's speed, as seen in the diagram below. This becomes an added factor in the player's balancing, as the player must manage their speed to navigate the area.



Luke Designer Notes!: The tilt currently depends on how LONG a direction is held, which feels quite unnatural and does not really depend on the speed at all.

Adjusting this to be more accurate may enhance the overall experience.

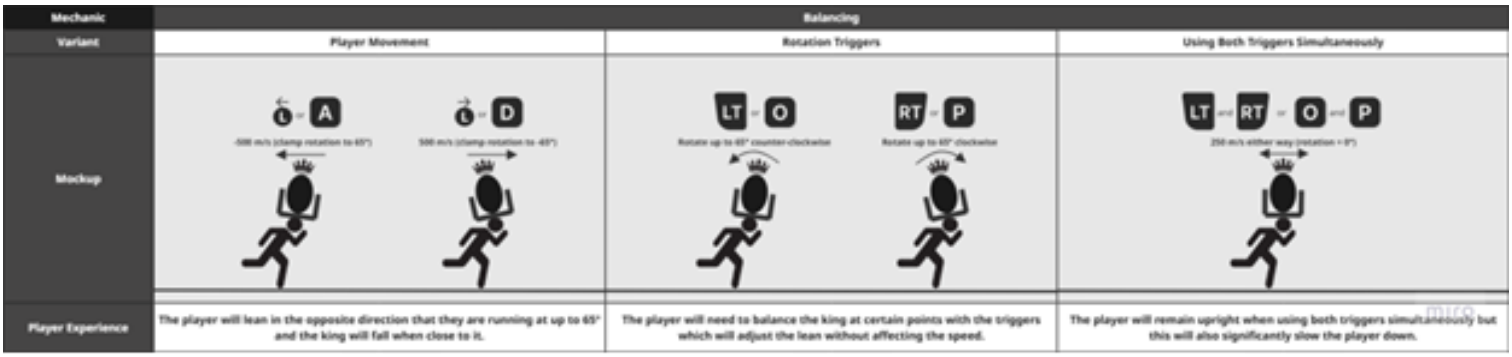
The back of the chair also makes it very easy to fully sprint and lean back without the egg falling out. Stopping after running for a while slingshot the character to standing, which feels unavoidable! Also, there is not much encouragement to use trigger tilt controls.

Ben's Notes: While making the balance work well in accordance with the player's speed on controller, how will this work with a keyboard if we include controls for that? As a key is a 0 or 1 input value, instead of a joystick which put through float values between 0 and 1.

Player Tilting

Luke Designer Notes: As of right now, when standing still, the tilt goes back and forwards around 20 degrees. The knight can tilt by moving and tilting, which gets it to about 45 degrees. Despite the extremes of this, the egg still holds on for a while. It feels the game should be a little more unforgiving in this situation!

When the player stops moving, they also snap back very fast to upright (this is



also pretty jarring, as this still happens even when the player slows down gradually)



Figure 1 Tilting the Knight with no force from movement



Figure 1 moving Backwards and tilting the Knight forwards



Figure 3 Moving forward and leaning backwards